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Expt 2 ASD&D

Aim - To containerise applications using Docker
LO - Containerize , orchestrate , and deploy applications using Docker , Docker Compose , Kubernetes and Cloud Infrastructure .
LO Mapped : LO4

Theory -

1. Introduction to Containerization

In traditional software deployment, applications often run into the **"it works on my machine"** problem. This happens because of differences in operating system versions, system libraries, configuration files, or dependency versions between a developer's local laptop and the production server.

Containerization solves this problem by bundling an application together with all of its dependencies, libraries, binaries, and configuration files into a single, isolated software package called a container. This ensures that the application runs identically across any computing environment.

2. What is Docker ?

Docker is a containerization platform that bundles an application alongside all its required dependencies, libraries, and configurations into a single, isolated package called a container. Unlike traditional virtual machines, containers share the host operating system's kernel, making them incredibly lightweight and fast to boot. This guarantees that your software runs identically across any machine, eliminating the classic "it works on my laptop" configuration problem.

3. Docker Architecture

Docker uses a client-server architecture. The three main pieces work together as follows:

[Docker Client] <----> [Docker Daemon (Host)] <----> [Docker Registry]
(CLI Commands) (Manages Containers) (Docker Hub)

1. **Docker Client (CLI):** The primary interface through which a user interacts with Docker. When a user types commands like `docker run`, `docker build`, or `docker rmi` into the terminal, the client sends these requests to the Docker Daemon.

2. **Docker Host (Docker Daemon / `dockerd`):** A persistent background process that listens for Docker API requests. It manages heavy-duty tasks such as building images, running containers, configuring networks, and managing storage volumes.
3. **Docker Registry:** A centralized storage repository for Docker images. **Docker Hub** is the default public registry where developers share pre-configured images for popular operating systems, databases, and language runtimes.

4. Core Components of Docker

An absolute understanding of Docker requires distinguishing between its three fundamental building blocks:

- **Docker Image:** A read-only, immutable template created from a Dockerfile. It contains the snapshot of the application code, libraries, and runtime environment. Images are organized as stacked layers to save storage space. An image cannot run by itself; it serves as the blueprint.
- **Docker Container:** A live, running, executable instance of a Docker image. It adds a thin, writeable layer on top of the read-only image layers, allowing the application to write files and execute processes in complete isolation.

```
C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41
```

docker --version - The command `docker --version` prints the currently installed Docker engine version, build details, and release info on your terminal. It's a quick sanity check to confirm that Docker is installed correctly on your system and to verify which release you are running.

```
C:\Users\15L>docker images
REPOSITORY                  TAG                                       IMAGE ID       CREATED         SIZE
sample-flask-app            v1                                        f7d64ef4ebbb   3 months ago    141MB
myredis                     version1                                  5c08b3fad205   3 months ago    140MB
redis                       latest                                    c234a24851ba   4 months ago    140MB
hello-world                 latest                                    1b44b5a3e06a   11 months ago   10.1kB
node                        latest                                    8bbb9d1bc633   2 years ago     1.1GB
nginx                       1.0                                       5682b6dc087e   2 years ago     169MB
ubuntu                      latest                                    ca2b0f26964c   2 years ago     77.9MB
ubuntu                      <none>                                    3db8720ecbf5   2 years ago     77.9MB
mysql                       latest                                    019814493c7a   2 years ago     632MB
mysql                       <none>                                    a88c3e85e887   2 years ago     632MB
httpd                       latest                                    2776f4da9d55   2 years ago     167MB
redis                       <none>                                    d1397258b209   2 years ago     138MB
docker/welcome-to-docker    latest                                    c1f619b6477e   2 years ago     18.6MB
getting-started             latest                                    8d8a2af2207f   3 years ago     265MB
docker/getting-started      latest                                    3e4394f6b72f   3 years ago     47MB
docker/desktop-git-helper   5a4fca126aadcd3f6cc3a011aa991de982ae7000   efe2d67c403b   4 years ago     44.2MB
```

docker images - The command `docker images` (or `docker image ls`) lists all the **Docker images** currently downloaded or built on your local machine.

```
C:\Users\15L>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
```

docker search - The command `docker search` searches the **Docker Hub registry** directly from your terminal for publicly available container images matching your search term.

```
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
ed819469700f: Pull complete
a3679419df18: Pull complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu
```

docker pull - The command `docker pull` downloads a specific container image (or set of images) from a remote registry—like **Docker Hub**—down to your local machine.

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

The command `docker ps` lists all **currently running Docker containers** on your system.

```
C:\Users\15L>docker run -it ubuntu /bin/bash
root@e446c46ef143:/#
```

The `docker run` command creates and starts a **new container instance** from a specified Docker image. It is the primary command used to execute applications inside Docker.

```
C:\Users\15L>docker start -i e446c46ef143
root@e446c46ef143:/# exit
exit
```

```
C:\Users\15L>docker stop e446c46ef143
e446c46ef143
```

```
C:\Users\15L>docker container stats
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
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CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
```

The command `docker stats` (or `docker container stats`) displays a **live, real-time stream of resource usage statistics** for your running containers.

```
C:\Users\15L>docker ps --all
CONTAINER ID   IMAGE                              COMMAND                  CREATED         STATUS                       PORTS                    NAMES
e446c46ef143   ubuntu                             "/bin/bash"              17 minutes ago  Exited (0) 4 minutes ago                              practical_austin
8ddb1e79e05f   ubuntu                             "/bin/bash"              20 minutes ago  Exited (0) 20 minutes ago                             distracted_ishiz
298abd2b1a9d   redis                              "docker-entrypoint.s…"   3 months ago    Exited (0) 3 months ago                               naughty_shirley
1a197c2bd0f5   redis                              "docker-entrypoint.s…"   3 months ago    Exited (255) 31 minutes ago  6379/tcp                 happy_galois
013ed608bab8   redis                              "docker-entrypoint.s…"   4 months ago    Exited (0) 4 months ago                               nifty_yonath
f9b579837c27   d1397258b209                       "docker-entrypoint.s…"   4 months ago    Exited (0) 4 months ago                               vigilant_knuth
06b8f6ece314   ca2b0f26964c                       "/bin/bash"              2 years ago     Exited (255) 4 months ago                             happy_cerf
7b8921bf4fb0   d1397258b209                       "docker-entrypoint.s…"   2 years ago     Exited (255) 3 months ago    6379/tcp                 jatinn
c782c1137761   node                               "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                jatingupta
ca1d141bfabd   node                               "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                reverent_willian
7638b03139a0   d1397258b209                       "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                jatin
d53dc9583439   d1397258b209                       "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                happy_engelbart
45c72eb69cd7   d1397258b209                       "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                keen_blackburn
b9b0d800f902   d1397258b209                       "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                priceless_jackso
a91dfde56008   d1397258b209                       "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                wizardly_sanders
87fe33760977   d1397258b209                       "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                epic_burnell
8823c8213441   ca2b0f26964c                       "/bin/bash"              2 years ago     Exited (0) 2 years ago                                AHMED
95a79a09eca1   ca2b0f26964c                       "/bin/bash"              2 years ago     Exited (0) 2 years ago                                festive_chatelet
aefcc9a16a00   ca2b0f26964c                       "/bin/bash"              2 years ago     Exited (0) 2 years ago                                festive_kare
2b923e117ec0   docker/welcome-to-docker:latest    "/docker-entrypoint.…"   2 years ago     Exited (0) 3 months ago                               welcome-to-docke
29dadd522f4b   docker/getting-started:latest      "/docker-entrypoint.…"   2 years ago     Exited (0) 2 years ago                                kind_bhaskara
cfb503a59d94   nginx:1.0                          "/usr/sbin/nginx -g …"   2 years ago     Exited (0) 2 years ago                                xyz
b730934a9b75   nginx:1.0                          "/usr/sbin/nginx -g …"   2 years ago     Exited (255) 2 years ago     0.0.0.0:9090->80/tcp     webserver
d2bc0c136fb2   httpd                              "httpd-foreground"       2 years ago     Exited (255) 2 years ago     80/tcp                   busy_edison
2e821f5cfabb   3db8720ecbf5                       "/bin/bash"              2 years ago     Exited (0) 2 years ago                                boring_diffie
0de78c8adee9   getting-started                    "docker-entrypoint.s…"   3 years ago     Exited (255) 3 years ago     0.0.0.0:3000->3000/tcp   frosty_tharp
38f4628b6bf4   docker/getting-started             "/docker-entrypoint.…"   3 years ago     Exited (255) 3 years ago     0.0.0.0:80->80/tcp       youthful_neumann
```

The command **`docker ps -all`** (or more commonly written as **`docker ps -a`**) lists **all containers** on your system—regardless of whether they are currently running, paused, or stopped.

```
C:\Users\15L>docker inspect e446c46ef143
[
    {
        "Id": "e446c46ef14302f1130fe4c5f489bba69c8c3fa8f036ee3ebe365c8557b32739",
        "Created": "2026-07-20T10:05:12.059193267Z",
        "Path": "/bin/bash",
        "Args": [],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 0,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-20T10:17:44.5401851477Z",
            "FinishedAt": "2026-07-20T10:17:54.144176276Z"
        },
        "Image": "sha256:de7345b16e942e22044c6ba053020ec85ae879984860a9918517d54eb6cef851",
        "ResolvConfPath": "/var/lib/docker/containers/e446c46ef14302f1130fe4c5f489bba69c8c3fa8f036ee3ebe365c8557b32739/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/e446c46ef14302f1130fe4c5f489bba69c8c3fa8f036ee3ebe365c8557b32739/hostname",
        "HostsPath": "/var/lib/docker/containers/e446c46ef14302f1130fe4c5f489bba69c8c3fa8f036ee3ebe365c8557b32739/hosts",
        "LogPath": "/var/lib/docker/containers/e446c46ef14302f1130fe4c5f489bba69c8c3fa8f036ee3ebe365c8557b32739/e446c46ef14302f1130fe4c5f489bba69c8c3fa8f036ee3ebe365c8557b32739-json.log",
        "Name": "/practical_austin",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
```

```
"HostConfig": {
    "Binds": null,
    "ContainerIDFile": "",
    "LogConfig": {
        "Type": "json-file",
        "Config": {}
    },
    "NetworkMode": "default",
    "PortBindings": {},
    "RestartPolicy": {
        "Name": "no",
        "MaximumRetryCount": 0
    },
    "AutoRemove": false,
    "VolumeDriver": "",
    "VolumesFrom": null,
    "ConsoleSize": [
        51,
        209
    ],
    "CapAdd": null,
    "CapDrop": null,
    "CgroupnsMode": "host",
    "Dns": [],
    "DnsOptions": [],
    "DnsSearch": [],
    "ExtraHosts": null,
    "GroupAdd": null,
    "IpcMode": "private",
    "Cgroup": "",
    "Links": null,
    "OomScoreAdj": 0,
    "PidMode": "",
    "Privileged": false,
    "PublishAllPorts": false,
    "ReadonlyRootfs": false,
    "SecurityOpt": null,
    "UTSMode": "",
    "UsernsMode": "",
    "ShmSize": 67108864,
    "Runtime": "runc",
    "Isolation": "",
    "CpuShares": 0,
    "Memory": 0,
    "NanoCpus": 0,
    "CgroupParent": "",
    "BlkioWeight": 0,
    "BlkioWeightDevice": [],
    "BlkioDeviceReadBps": [],
    "BlkioDeviceWriteBps": [],
```

```
C:\Users\15L>docker logs  e446c46ef143
root@e446c46ef143:/# ^C
root@e446c46ef143:/# exit
exit
root@e446c46ef143:/# exit
exit
```

The command **`docker inspect`** retrieves detailed, low-level configuration and runtime metadata about Docker objects (containers, images, volumes, networks, or plugins) in **JSON format**.

```
C:\Users\15L>docker rm e446c46ef143
e446c46ef143
```

```
C:\Users\15L>docker build -t my-app .
[+] Building 0.0s (1/1) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 2B
ERROR: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory

View build details: docker-desktop://dashboard/build/default/default/sbqyz7djgxok8cb2bd7yv1ob7
```

```
C:\Users\15L>docker build -t my-app --no-cache .
[+] Building 0.0s (1/1) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 2B
ERROR: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory

View build details: docker-desktop://dashboard/build/default/default/zksahc9309kw9t8o4dczh1upt
```

The command **`docker build`** creates a new custom **Docker image** from a text file called a **Dockerfile** and a set of local files (known as the *build context*).

```
C:\Users\15L>docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: mysql@sha256:ff5ab9cdce0b4c59704b4e2a09deed5ab8467be795e0ea20228b8528f53fcf82
deleted: sha256:a88c3e85e887d78c6c16ee6a0a6297e09bd573193918a08f269a942ddad77856
deleted: sha256:a47e3f5184188d0ded58167f81ed399bda838b37716d6a0d9fb538ae64371767
deleted: sha256:750ea5242272cdbf344d0af35c4eff64c7b0296491bf9747214d4012c4feb391
deleted: sha256:2f17d48c81db0969179826d20d81e1cd333676a714e293a19a6fdaa2d57b1f27
deleted: sha256:bea3f162c4bd2492d6cfce9f715c60a61020205e82647343034c33c2d2c1c720
deleted: sha256:d3ad25eecba2cc8f5d837c5436fd4ce2230c51c6a6b206125b9dba76f83bd02f
deleted: sha256:8fb5524227094d3e8963e8bb93af8835a5928f34a072e2272aaf330b00fc3a58
deleted: sha256:3dcdcc4db1094b30a257b6feb414db6149934171333bc8c6ea543578021862ac
deleted: sha256:706b673eebeed9bc92da642eee3a1f5ea076e149b98491624ce7436ee40d115d
deleted: sha256:ed862c4f3420f8cdf5fb1160305a714585ffd0cc739c6f6306c750e82d50cd75
deleted: sha256:55d310cc1793fd699f7389585b86d5ff0270ad8f516f81db295e51e12f6d0b5b

Total reclaimed space: 632.3MB
```

The command **`docker image prune`** cleans up disk space by deleting **dangling** (untagged) Docker images from your local system.

Conclusion - LO4 mapped successfully.